

Program Name: Engineering

Level: Degree

Branch: All

Course / Subject Code : BE02000011 Course

/ Subject Name : Mathematics – 2

Important Topics for GTU Mathematics-2 Exam

Module 1: Matrices

1. Rank of a Matrix using REF/RREF.
2. Inverse of a Matrix using the Gauss-Jordan method.
3. System of Linear Equations solved by row operations.
4. Properties of Symmetric, Skew-Symmetric, and Orthogonal Matrices.
5. Eigenvalues and Eigenvectors, and Diagonalization of matrices.
6. Cayley-Hamilton Theorem for the inverse of a matrix.

Module 2: First-Order ODEs

1. Exact Differential Equations and Integrating Factors.
2. Linear and Bernoulli Equations.
3. Equations solvable for p , x , y and Clairaut's type equations.

Module 3: Higher-Order ODEs

1. Linear Differential Equations with constant and variable coefficients.
2. Euler-Cauchy Equations.
3. Method of Undetermined Coefficients and Variation of Parameters.
4. Power series solutions at ordinary points.

Module 4: Complex Variables (Differentiation)

1. Cauchy-Riemann Equations and analyticity.
2. Harmonic Functions and their conjugates.
3. Analyticity of exponential, trigonometric, and logarithmic functions.

Module 5: Complex Variables (Integration)

1. Contour Integration and the Cauchy-Goursat Theorem.
2. Laurent's Series and classification of singularities.
3. Cauchy Residue Theorem and applications.

Focus on these areas as they frequently appear in past papers.

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